Understanding the Problem

How did you approach understanding the challenge?

I first read through the exercise and identified what the program was required to have. I then broke the problem into smaller parts, input, process, and output. In each section I worked to optimize the code, and I added extra features that weren't required, such as error handling. I spent the majority of my time refining the masteries, as they matter more than the skilbuilders.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I initially found utilizing different methods difficult, but as I practiced I began to understand them more and I resolved my confusion.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes I broke the program down into steps. First I planned how to get the user's input, then I planned what calculations the computer would perform with that information, and finally I planned what the computer would output.

How did you decide on the tools, data structures, or algorithms to use?

I used the tools that I believed were best for the given task. I used loops for error handling, booleans for true or false statements, and simple integers to perform calculations. I also used different methods to perform larger tasks, so that my code would remain simple and straightforward.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I had already planned out each section of the code in smaller pieces, so when I wrote it I took inspiration from my planning and worked on each section individually while ensuring that they all worked together in harmony.

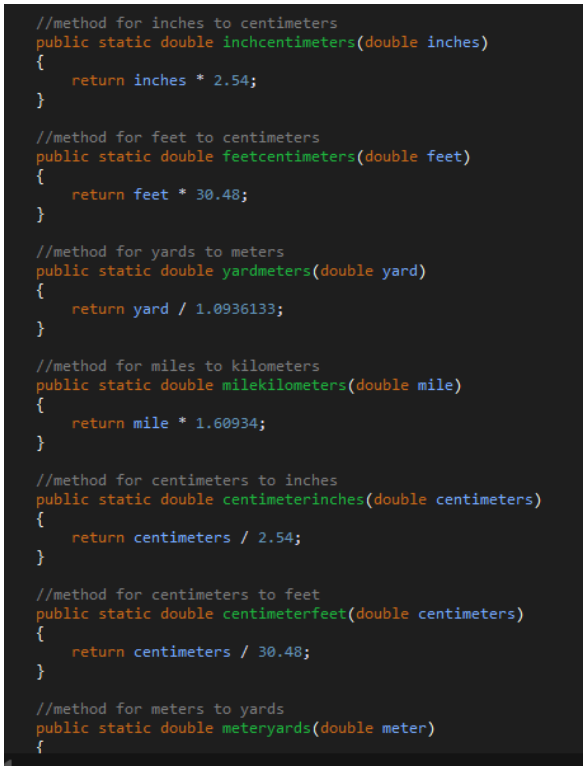
How did you test your solution along the way to make sure it was working?

I would run the program many times throughout the coding process. I wanted to ensure that each new line of code performed its task correctly before continuing.

Overcoming Challenges

What part of the problem was the most difficult for you?

For me the most challenging part was utilizing multiple different methods throughout my code. I had to learn how different methods worked, how to define them, and how to make them perform and return calculations using my desired values.



```
//method for inches to centimeters
public static double inchcentimeters(double inches)
{
    return inches * 2.54;
}

//method for feet to centimeters
public static double feetcentimeters(double feet)
{
    return feet * 30.48;
}

//method for yards to meters
public static double yardmeters(double yard)
{
    return yard / 1.0936133;
}

//method for miles to kilometers
public static double milekilometers(double mile)
{
    return mile * 1.60934;
}

//method for centimeters to inches
public static double centimeterinches(double centimeters)
{
    return centimeters / 2.54;
}

//method for centimeters to feet
public static double centimeterfeet(double centimeters)
{
    return centimeters / 30.48;
}

//method for meters to yards
public static double meteryards(double meter)
{
```

How did you handle moments when you felt stuck or unsure of what to do next?
I looked at the code carefully, examined the error messages, and conducted research to learn why it wasn't performing its task correctly and how I could fix it. I also looked back on my previous code to see what I was doing differently.

Learning
Was there anything you learned that you think will help you with future challenges?
I learned how much is truly possible with Java through its math function. There are so many different methods that can be programmed to perform a variety of tasks, and it makes me realize how much Java can truly do.